

Reactions of Organic Compounds

Reaction types, products, tests and yields

Name _____ Class _____ Date _____

REAGENTS AND DATA

Addition: H_2/Ni · Br_2 · HBr · $\text{H}_2\text{O}/\text{H}^+$ · Substitution: Cl_2/UV · $\text{OH}^-(\text{aq})$ · Elimination: conc. H_2SO_4 , heat · Oxidation: acidified $\text{K}_2\text{Cr}_2\text{O}_7$ (orange $\rightarrow$ green)

Molar masses (g/mol): C_2H_4 28.05 · $\text{C}_2\text{H}_5\text{OH}$ 46.07 · CH_3COOH 60.05

Show units on every quantity in every step.

A. Understanding

1. Define, with one example each: (a) addition (b) substitution (c) elimination (d) oxidation (e) condensation.

B. Reaction types and products

2. Classify each reaction: (a) $\text{CH}_2=\text{CH}_2 + \text{H}_2 \rightarrow \text{CH}_3\text{CH}_3$ (b) $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$ (c) $\text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$ (d) ethanol $\rightarrow$ ethanoic acid (e) ethanoic acid + methanol $\rightarrow$ methyl ethanoate + water (f) $\text{CH}_3\text{CH}_2\text{Br} + \text{OH}^- \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{Br}^-$

3. Name the organic product: (a) but-2-ene + H_2 (Ni) (b) but-2-ene + Br_2 (c) chloroethane + OH^- (aq) (d) propan-1-ol heated under reflux with acidified dichromate (e) butan-2-ol warmed with acidified dichromate

C. Tests and selectivity

4. Describe a chemical test, with the observations, to tell apart: (a) hexane and hex-1-ene (b) propan-1-ol and 2-methylpropan-2-ol.

5. Propan-1-ol can be oxidized to propanal or to propanoic acid. How do the conditions decide which product forms?

6. Propene reacts with HBr . (a) Name the two possible products. (b) Which is the major product, and why?

D. Calculations

7. Ethanol is made industrially by $\text{CH}_2=\text{CH}_2 + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{CH}_2\text{OH}$. In one run, 14.0 kg of ethene gives 21.2 kg of ethanol. Calculate the theoretical yield and the percent yield.

8. 9.21 g of ethanol is completely oxidized to ethanoic acid. What mass of ethanoic acid forms?

E. Challenge

9. Starting only from ethene, water and the reagents in the data box, plan a synthesis of ethyl ethanoate. Give each step, its reagents and conditions, and the type of reaction.